

Shaan Patel

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Education

University of Texas at Arlington

Ph.D. in Physics and Applied Physics 2025

- Published 7 publications and delivered research presentations at 3 physics conferences

M.S. in Physics, GPA: 4.00 2025

Georgia Institute of Technology

B.S. in Physics, Highest Honor 2019

Experience

Computational Astrophysics Researcher | UT Arlington 2021 – 2025

- Conducted 50,000+ numerical simulations in Python with the Exoplanets group to investigate orbital dynamics, improving runtime efficiency by 80%
- Encoded mathematical formulations into executable code, enabling efficient large-scale numerical analysis and reducing computation time by 90%
- Analyzed and visualized large-scale datasets comprising millions of data points, identifying key dynamical patterns

Physics Research Intern | SLAC National Accelerator Lab 2019

- Worked with the Cryogenic Dark Matter Search (CDMS) team to test wiring and readout electronics for the He-3/He-4 dilution refrigerator

Lead Undergraduate Researcher | Georgia Institute of Technology 2017 – 2019

- Led a research team in the Numerical Relativity Research Group simulating binary black hole dynamics on HPC clusters, analyzing high-dimensional gravitational wave data for LIGO research

Investment Analyst | Georgia Institute of Technology 2016 – 2017

- Conducted research, financial modeling, and valuation analyses for the Student Investments Committee to manage and pitch equity investments for a multi-million-dollar student-run endowment

Projects

AuroraNet: Capstone Aurora Prediction Project

- Engineered an end-to-end time-series pipeline in Python (pandas, NumPy) to clean 24 years of space weather data, training PyTorch LSTM models to achieve an RMSE of 0.642
- Deployed real-time inference endpoints via FastAPI and built an interactive Streamlit dashboard, containerizing the app with Docker and logging experiments using Weights & Biases (W&B)

DefCoordML: NFL Play Prediction

- Developed ML models to predict NFL offensive plays (pass vs. rush) using historical game data
- Achieved 75% accuracy and 0.82 ROC AUC through feature engineering and model tuning

Skills

Languages

Python, SQL, Bash

Data Science & Visualization

NumPy, pandas, SciPy, matplotlib, seaborn, Jupyter

Machine Learning

PyTorch, Scikit-learn, Weights & Biases (W&B)

Deployment & Engineering

Docker, FastAPI, Streamlit, Git, Linux, PyTest